

# A utility-level analysis of residential electricity use: Governance, pricing, and incentive programs in the Southeastern U.S.

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## ABSTRACT

This study examines utility-level variation in annual residential electricity consumption across Georgia and North Carolina, focusing on utility governance, electricity pricing, service-territory characteristics, and utility-run incentive programs. We construct a hand-collected inventory of 699 residential energy programs across 193 electric utilities and merge it with utility-level data, such as electricity consumption, rate, energy burden, climate, demographics and housing conditions, and EV charging-station data. The analytic sample includes 105 utilities with complete consumption and covariate data. Our results show that each additional EV program is associated with approximately 310 kWh (2.5%) higher annual per-household use, even after controlling for public EV charging-station density. Cooperative utilities initially exhibit higher household consumption than municipal utilities, but this difference loses statistical significance after accounting for program offerings and demographic factors. A one-cent/kWh higher electricity rate is associated with a 227 kWh (1.8%) to 334 kWh (2.7%) reduction in annual per-household electricity consumption, which may reflect both efficiency-driven conservation and energy-limiting behaviors. Prevalence of energy burden is associated with more electricity use after controlling for income, suggesting structural inefficiency in high-burden utility territories. These findings indicate that utility incentive programs should be paired with managed charging, demand response, targeted efficiency investment, and time-of-use pricing.

## 1. Introduction

Residential energy consumption is a critical component of energy demand in the United States, accounting for over 20% of total electricity use nationally (US EIA, 2023). Understanding what drives variation in household energy use is increasingly important for shaping equitable and effective energy policy, as utilities, regulators, and policymakers strive to reduce greenhouse gas emissions, promote energy efficiency, and alleviate the disproportionate energy burdens faced by low-income communities. While past research has extensively investigated household-level energy use, much of it has relied on micro-level data,

such as individual consumption records, smart meter data, or survey responses (e.g., Lee and Song, 2022; Brown et al., 2023; Antonopoulos et al., 2019), limiting the ability to inform utility-scale policy interventions.

This study advances the literature by shifting the analytical lens from the individual household to the utility level, focusing on the Southeastern U.S. states of Georgia and North Carolina. Utilities in these two states operate under varying governance structures (i.e., investor-owned, municipal, and cooperative) and serve diverse populations with heterogeneous demographic, socioeconomic, and housing characteristics. These differences, coupled with divergent programmatic

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offerings such as energy efficiency (EE) rebates, renewable energy (RE) programs, or electric vehicle (EV) incentives, present a unique opportunity to assess how utility-level characteristics relate to aggregated residential electricity use.

Utility-level analysis is vital for three reasons. First, utilities are central actors in shaping energy consumption patterns through the rates they set, the programs they offer, and the technologies they promote. In many cases, utilities also serve as the primary conduit for state and federal energy policy implementation. Second, aggregating data to the utility level better reflects the actual jurisdictional boundaries within which energy programs and investments are administered. Finally, examining variation across utilities enables researchers and policy-makers to detect structural patterns that may be obscured in household, census tract and county-level studies, including how governance structure and program offerings shape energy use in aggregate.

This study draws on original data compiled from 193 electric utilities in the states of Georgia and North Carolina. We match these utilities with data from both proprietary sources ([PowerOutage.com](https://www.poweroutage.com), 2026) and public sources, including the U.S. Department of Energy (DOE)'s Energy Information Administration (EIA), the Low-Income Energy Affordability Data (LEAD) Tool ([United States Department of Energy Office of State and Community Energy Programs](https://www.energy.gov/eere/lead), n.d.), Census Bureau, and PRISM Group at Oregon State University. Our final dataset tracks 105 utilities with complete consumption, utility-level characteristics, and demographic data available.<sup>2</sup>

With the comprehensive datasets, we assess how average annual household electricity consumption across utility service territories relates with: (1) climate, demographic, and housing characteristics (e.g., cooling and heating degree days, education levels, prevalence of electric heating and number of rooms); (2) economic constraints and structural conditions, such as electricity rates and prevalence of energy burden; (3) utility governance models, comparing investor-owned utilities (IOUs) to municipal utilities (MUNIs) and cooperatives (COOPs); and (4) programmatic interventions, namely, availability of utility-run programs promoting EE, RE adoption, and EV usage.

Our approach is grounded in prior research that identifies key determinants of household energy use (e.g., housing characteristics, income levels, climate, and behavioral patterns) but we apply this knowledge to a new level of analysis: electric utility service territories. This aggregated approach is valuable in regions like the Southeast, where energy poverty is widespread ([Chen et al., 2022](#); [Brown et al., 2020](#)), and where utilities vary widely in size, mission, and regulatory oversight.

This study contributes to the literature in three ways. First, it shifts the empirical unit from households, census tracts, and counties to electric utility service territories. As mentioned above, this matters because utility rates, incentive programs, and governance structures are designed and implemented at the utility scale, not at the census-tract or county scale. Second, the study introduces a hand-collected inventory of residential energy-related programs across all operating electric utilities in Georgia and North Carolina, including municipal and cooperative utilities that are often underrepresented in centralized program databases. Third, the analysis quantifies the magnitude of associations between program portfolios, governance structures, pricing, and residential electricity use. The central contribution is therefore not the directional finding that larger homes or electric heating are associated with higher consumption. It is the evidence that utility program portfolios, especially EV incentives, are meaningfully associated with aggregate residential demand and that apparent ownership differences attenuate once program offerings and service-territory characteristics

are considered.

These contributions are especially relevant for policy because utilities are increasingly expected to support electrification, energy affordability, and decarbonization simultaneously. Utility-level evidence can help regulators and utility managers evaluate whether program portfolios are aligned with demand management and equity goals, and whether EV, RE, and EE incentives should be coordinated with pricing, managed charging, and targeted efficiency investments.

## 2. Literature review

Utilities in the United States play a key role in affecting their customers' energy consumption through their policies, decisions and programs, such as energy efficiency (EE) programs. Utilities are of particular interest because their programs directly interact with customers and they are best positioned to motivate end-users to reduce consumption. Thus, it is important to measure utility performance by evaluating ratepayers' energy consumption at the utility level and understanding the contributing factors. However, existing research has primarily focused on analyzing individual household's energy consumption, lacking understanding of the overall pattern on the utility level. Still, findings from previous studies provide valuable information on the factors affecting energy use in buildings. Understanding these factors is useful for developing solutions to improve EE and alleviate energy burden for families under economic stress ([Scheier and Kittner, 2022](#)).

Below, from the relevant prior literature, we begin with a summary of these factors affecting residential building energy consumption, including home activity, climate, structure, efficiency, price, and demographics. Note that the behavioral factor is not discussed extensively because (1) it is out of the scope of the current study; and (2) occupant behaviors affect home energy use indirectly through the choices and decisions about home appliances, electric devices, and the physical characteristics of buildings and systems ([Sanquist et al., 2012](#); [Steemers and Yun, 2009](#)).

**Home Activity Factors.** Home activity directly determines energy consumption, and is influenced by the number of household members and occupancy patterns across different hours, weekdays, and weekends ([Berg et al., 2024](#); [Lee and Song, 2022](#)). Families with children, especially older kids, tend to use more energy than others ([van den Brom et al., 2018](#)). More broadly, the number of household members and their sociodemographic characteristics affect cooking, lighting, and the use of building equipment and appliances.

**Climate Factors.** Another direct determinant of home energy use is weather and climate. In particular, heating and cooling degree days affect large end-uses, such as air-conditioning, and space and water heating ([Cui et al., 2024](#); [Moustris et al., 2015](#)). Degree days are useful to capture regional variations, and it provides an average estimation of the long-term weather pattern ([Şen, 2022](#); [Xu et al., 2016](#)).

**Housing Characteristics.** Important physical housing characteristics include building type, building size (floor area, number of rooms), year of construction, number of windows, and number of air-conditioners ([Antonopoulos et al., 2019](#); [Cui et al., 2024](#); [Lee and Song, 2022](#)). Building equipment and large appliances, such as the type of heating, ventilation, and air-conditioning (HVAC) equipment, water heaters, refrigeration, dishwashers, clothes washers and dryers, are also determinants of energy use, because they represent end-uses that consume the most energy in residential buildings ([Lee and Song, 2022](#)).

**Efficiency Factors.** The intensity factor of buildings, often referred to as energy efficiency, is measured as energy consumption per square footage of floor area, which is mostly determined by the thermal performance of the building envelope and the efficiency of building equipment and appliances. Owned homes tend to have higher efficiency than rented residences, while higher education of occupants is known to lead to higher efficiency ([Kavousian et al., 2015](#)). Significant energy savings can be reached through improvements such as efficient boiler,

<sup>2</sup> EIA-861 "Annual Electric Power Industry Report" does not report residential electricity consumption data for 88 electric utilities excluded from the analysis. However, we show that these excluded utilities are very small ones, making only 4% of total population served in the states.

cavity wall insulation, and other retrofits of the building envelope (Adan and Fuerst, 2016; Desvallées, 2022). Residential building EE can also be affected by many other factors, such as the use of home energy management systems (HEMS). These systems can save energy by optimizing the use of home equipment and appliances (Jin et al., 2017; Mahapatra and Nayyar, 2022). EE can lead to additional benefits on the aggregated level, such as cooling the surrounding neighborhoods, while increasing buildings' heat resilience, leading to health and environmental benefits (Baniassadi et al., 2022).

**Electricity Price.** Electricity use in residential buildings also react to external market factors, such as price. Previous studies suggest that households in high price regions tend to have lower energy consumption levels (Sanquist et al., 2012). However, home energy demand is typically inelastic with respect to price (Fabra et al., 2021; Lanot and Vesterberg, 2021; Trotta et al., 2022). That is, energy consumption in residential buildings is quite stable, which does not vary to the same extent in response to other price variations. This could be caused by multiple reasons, such as “indifference” or lack of knowledge on energy cost, the habit of keeping a routine in appliance usage, and lack of information for energy conservation or efficiency improvement. Yet, households are sensitive to appliance prices, and high prices present a significant hurdle to the adoption of efficient appliances (Yohanis, 2012).

**Demographic Factors.** The age and employment status of the household head is known to be marginally related with heating consumption (Lee and Song, 2022). Time trend of the U.S. residential energy consumption increases with age, primarily caused by growing housing size over the years (Estiri and Zagheni, 2019). Some studies find that high education is associated with energy conservation (Kavousian et al., 2015), but others find no meaningful difference (Trotta, 2018). In less developed regions, though, high education could be related with high energy consumption (Wang et al., 2020). Household energy consumption is often examined for different income groups. Income is often positively correlated with household energy consumption (Sapci and Considine, 2014), especially in less developed region (Wang et al., 2020). In more developed settings, the estimated effect of income could be very small, with estimated contribution of only 1%, representing a smaller influence than climate and physical housing factors (Sanquist et al., 2012). The impact of income on residential energy consumption is primarily through the influence of choices in physical housing characteristics, where high income is correlated with high floor area, and more heated and air-conditioned rooms, which subsequently lead to high home energy use.

**Prevalence of Energy Burden.** Many low-income families find it challenging to pay their energy bills, a problem often referred to as energy burden. A study finds that 16% of American families face energy burden with more than 6% of their income spent on energy bills (Scheier and Kittner, 2022). The lowest-income group has an energy burden as high as 14% in summer, a severe problem in Arizona given future climate impacts on increased electricity use in hot climate regions (Jones et al., 2023). This burden disproportionately may affect certain vulnerable households, such as single-mother families with children (Brown et al., 2023). Since energy burden results from low-income, poor housing conditions, and low energy efficiency, alleviating energy poverty requires coordinated housing, energy, and health policies (Brown et al., 2020; Hernández and Bird, 2010a; Chen et al., 2022a, 2022b, 2024).

### 3. Research questions

Based on the aforementioned body of knowledge on household-level energy use, this study asks the following research questions to examine the relationships between home electricity consumption and the home activity, climate, structure, efficiency, and price factors, along with energy burden and population demographics within utility territories. The behavioral factor is not included because it affects energy consumption indirectly and our unit of analysis is electric utility service

territories.

**RQ1:** *What factors, such as home activity, climate, structure, and demographics at the utility level predict the average home electricity use of ratepayers?*

**RQ2:** *How do utility-level prevalence of energy burden and electricity price predict the average home electricity use of ratepayers?*

It is also worth investigating whether utility ownership could affect the aggregated measure of energy consumption of residential customers. Compared with investor-owned utilities (IOUs), municipal and cooperative utilities are uniquely positioned to promote demand-side management (DSM) resources for their customers (Wilson et al., 2008). Research also suggests that cooperatives or IOUs tend to disconnect their electric customers more often than municipal utilities (Memmott et al., 2023, 2024).

The type of utilities may reveal the differences in their customers' energy consumption because different patterns in home energy use are observed among urban and rural dwellers. Rural households have higher electricity consumption than city and town households, primarily driven by more frequent use of appliances, such as clothes washers and dryers, and dishwashers, as well as higher heating degree days. The lower electricity consumption by urban household is due to lower air-conditioned square footage and lower usage of home appliances (Sanquist et al., 2012). Research also finds that electricity use is lower among homes in more dense developments (Antonopoulos et al., 2019), as a result of the canopy effect of urban buildings and trees (Ravazdeh and Rivers, 2025; Tadeu et al., 2021). Dense development could also allow for lawn area to reduce surface heating in summer (B. Wilson, 2013).

On the other hand, the urban heat island effect could lead to increased demand for air-conditioning in summer, leading to increased residential energy cost (Roxon et al., 2020; Yang et al., 2022). With IOUs typically serving urban residents, compared with municipal utilities in small cities and towns and rural co-ops operating in rural areas, our second research question is proposed to examine the implications of the distinctive utility governance models.

**RQ3:** *Do IOUs perform differently from municipal utilities and rural cooperatives in the average home energy use of ratepayers?*

It is critical for utility companies to help families improve energy efficiency and encourage conservation, which is helpful for families experiencing high energy burden (Chen et al., 2021; Hernández and Bird, 2010b; Jones et al., 2023). Some utilities have been implementing programs to motivate customers to adopt efficiency improvements at home, including rebate programs to encourage the switch to high-efficient appliances, energy audits to provide information and suggestions, and financial incentives for insulation projects.

For example, the DSM programs by certain utilities in Florida are estimated to save 11% energy for heat pump rebates, 4% for attic insulation rebates, and 3% for energy audit (Taylor et al., 2014). Additionally, utility programs should target the least-efficiency customers to reach the most savings (Taylor et al., 2014). Facing an increasing cooling need caused by climate change, a study in Arizona appeals for intentional consideration of energy affordability of the vulnerable population in boosting energy-efficient technology adoption (Jones et al., 2023).

On the other hand, the adoption of renewable energy and EVs may increase electricity consumption. For example, studies have detected the solar rebound effect, an increase in energy use among adopters of solar rooftop PV panels (Beppler et al., 2023; Deng and Newton, 2017). The rebound effect could be driven by decreased energy bills, or simultaneous adoption of EVs (Beppler et al., 2023). Additionally, a rebound effect is also observed for energy efficiency, which originates from behavior where people use more energy (directly and indirectly) when

they adopt efficient technologies (Greening et al., 2000; Sorrell et al., 2009). The rebound effect can greatly diminish the effectiveness of energy efficiency programs, with an effect size of 27% among homeowners and 41% among tenants for heating (Aydin et al., 2017) and an overall effect of 56–80% for the residential sector (Orea et al., 2015).

We aim to quantify how the availability or offerings of utility programs predicts residential electricity usage by jointly considering programs for EE, RE, and EVs. Specifically, the last research question we ask is:

**RQ4:** *How does the availability or offerings of utility programs (EE, RE, and EVs) correlate with the average home energy use of ratepayers?*

By answering these research questions, this study contributes to the knowledge gap of evaluating utilities' overall performance using their residential customers' average electricity consumption.

#### 4. Data and methods

This study uses annual average household electricity consumption per utility for the dependent variable. Using the information about total electricity sales and the number of residential customers from the [United States Energy Information Administration, 2024](#) Form EIA-861 (i.e. Annual Electric Power Industry Report), we derive household electricity consumption. In addition to household electricity consumption, we use per capita electricity consumption to examine whether our results vary when shifting our focus from household to individual-level.<sup>3</sup> The results remain virtually the same, and we report the findings with household measures. To be clear, our unit of analysis is electric utilities. We collected incentive program data for all 193 utilities in Georgia and North Carolina from their respective websites and documents, but relied on EIA data to derive per-household energy consumption. Since EIA data does not cover all utilities, those with missing consumption data are excluded. The final analysis captures 105 of the 193 utilities.

Table 1 provides a detailed overview of all independent and dependent variables used in our analysis at the utility level, along with their corresponding data sources. Table 2 presents the summary statistics for these variables.

##### 4.1. Geographic scope of the study

Georgia and North Carolina offer a compelling comparative context for analyzing utility-level residential electricity consumption due to their institutional diversity, demographic variability, and regional relevance within the Southeastern United States. The two states collectively host nearly 200 electric utilities, spanning investor-owned (IOUs), municipal (MUNIs), and cooperative (COOPs) governance structures, which allows for systematic comparison of how utility type and energy incentive program availability shape household electricity use.

This variation is salient in a region where energy poverty remains prevalent and the burden of energy costs disproportionately affects low-income and marginalized households (Brown et al., 2020; Scheier and Kittner, 2022). The Southeast, including Georgia and North Carolina, is also characterized by relatively high electricity consumption and low energy efficiency investments compared to other U.S. regions, making it a strategic site for evaluating how utilities influence household demand and energy equity (Drehobl et al., 2020; Brown et al., 2023).

The two states further represent contrasting regulatory environments. North Carolina has pursued more proactive policies in areas such as renewable energy development and electric vehicle infrastructure, driven in part by its Renewable Energy and Energy Efficiency Portfolio

**Table 1**

Data sources.

| Variables                   | Description                                                                                                                                | Data Source                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| White                       | Percent of non-Hispanic white                                                                                                              | Census ACS                                 |
| Higher Education            | Percent of Bachelor or higher                                                                                                              | 5-year Estimates                           |
| Income                      | Median household income                                                                                                                    |                                            |
| Renter                      | Percent of renters                                                                                                                         |                                            |
| Built After 2010            | Percent of houses built after 2010                                                                                                         |                                            |
| Median Year Built           | Median built year of the structures                                                                                                        |                                            |
| Single Unit                 | Percent of structures with single unit                                                                                                     |                                            |
| Median Number of Rooms      | Median number of rooms within the structure                                                                                                |                                            |
| Electricity for Heating     | Percent of households using electricity as heating source                                                                                  |                                            |
| State                       | Utilities in GA are coded as 0 and utilities in NC are coded as 1                                                                          |                                            |
| Urban Area                  | Percent of urban areas                                                                                                                     | Census Urban Area TIGER/Line Shapefiles    |
| Energy Burdened Population  | Percent of households spending more than 6% of their annual gross household income on energy costs                                         | DOE LEAD (Low-Income Energy Affordability) |
| EV Charging Station Density | Number of EV charging stations per 10,000 population                                                                                       | DOE                                        |
| Electricity Rate            | Average electricity rates from 2017 to 2022                                                                                                | PowerOutage.com                            |
| Heating Degree Days         | Degree days below 15 °C                                                                                                                    | PRISM                                      |
| Cooling Degree Days         | Degree days above 30 °C                                                                                                                    |                                            |
| Utility Ownership Structure | A set of binary variables indicating ownership structure for utility: Investor-owned (IOU), Cooperative (COOP), and Municipal-owned (MUNI) | Utility Websites                           |
| EE Incentive Program        | Number of energy efficiency incentive programs                                                                                             |                                            |
| RE Incentive Program        | Number of renewable energy incentive programs                                                                                              |                                            |
| EV Incentive Program        | Number of electric vehicle incentive programs                                                                                              |                                            |

**Table 2**

Summary descriptive statistics for key variables.

| Variables                       | Mean   | SD     | Min    | Max     |
|---------------------------------|--------|--------|--------|---------|
| Per Household Consumption (kWh) | 12,326 | 1634   | 8563   | 15,371  |
| Per Capita Consumption (kWh)    | 4882   | 621    | 3413   | 6626    |
| Electricity Rate (\$/kWh)       | 0.13   | 0.01   | 0.10   | 0.16    |
| Non-Hispanic White (%)          | 60     | 18     | 8      | 98      |
| Higher Education (%)            | 25     | 10     | 13     | 65      |
| Income (\$)                     | 57,752 | 17,478 | 27,889 | 123,724 |
| Single Unit (%)                 | 71     | 9      | 44     | 91      |
| Renter (%)                      | 32     | 12     | 12     | 69      |
| Built After 2010 (%)            | 8      | 5      | 1      | 28      |
| Median Built Year               | 1985   | 8      | 1963   | 2002    |
| Median Number of Room (count)   | 6      | 1      | 5      | 8       |
| Prevalence of Energy Burden (%) | 35     | 14     | 10     | 96      |
| Electric Heating (%)            | 65     | 17     | 21     | 93      |
| Urban (%)                       | 31     | 37     | 0      | 1       |
| CDD                             | 45     | 22     | 0      | 94      |
| HDD                             | 1100   | 310    | 549    | 2156    |
| EE incentive programs (count)   | 2      | 2.0    | 0      | 9.0     |
| RE incentive programs (count)   | 1.1    | 1.1    | 0      | 4.0     |
| EV incentive programs (count)   | 0.4    | 0.7    | 0      | 3.0     |

Notes: CDD = Cooling Degree Days. HDD = Heating Degree Days. EE = Energy Efficiency.

RE = Renewable Energy. EV = Electric Vehicle.

Standard (REPS). In contrast, Georgia remains largely dominated by vertically integrated IOUs with centralized control over rate structures and generation planning (Wilson et al., 2008).

<sup>3</sup> Per capita electricity consumption was calculated by dividing household electricity consumption by the average household size obtained from the US Census Bureau's American Community Survey (ACS).



Despite these policy differences, both states share similar climatic conditions, marked by long, hot summers and mild winters, which helps control for temperature-driven variation in electricity demand. Together, Georgia and North Carolina offer a heterogeneous yet geographically and climatically consistent setting for examining how utility governance, pricing, and programmatic interventions relate to household electricity consumption patterns.

## 4.2. Variables and measurements

### 4.2.1. Socioeconomic factors

This study used sociodemographic variables from the U.S. Census Bureau's ACS, which includes race, income, education, and occupancy type. Building characteristics, such as age and structure, and main fuel sources for heating are known to influence energy consumption (Aksoezen et al., 2015). We therefore included building age, the percentage of single unit structures, and the proportion of households using electricity as their heating source.

### 4.2.2. Prevalence of Energy Burden

This study included energy burden as an exploratory variable. A household is considered energy-burdened if it spends more than 6 % of its gross annual income on residential energy expenses, following U.S. DOE's guidance (DOE Office of State and Community Energy Programs, n.d.).<sup>4</sup> To construct this variable, we use tract-level data from the LEAD tool, which provides average annual residential energy costs, including electricity, utility gas, bottled gas, fuel oil, wood, and other fuels. Median household income for each tract comes from the American Community Survey.

For every tract, we compared the average annual energy cost with 6 % of median income to estimate the proportion of households whose energy-cost-to-income ratio exceeds the 6 % threshold. We then aggregated these tract-level estimates to the utility service territory by computing the percentage of households within each utility that are energy burdened.

### 4.2.3. Electric vehicles charging infrastructure

To account for EV-driven electricity demand, we included the number of EV charging stations collected from Alternative Fuels Data Center from DOE (DOE, n.d.). While this data does not contain information about home chargers, we included the number of EV charging stations as a proxy of EV-driven electricity demand in our analysis. We considered this proxy to reflect the number of EV users in a given utility service territory.

### 4.2.4. Urban area

This study incorporated the percentage of urbanized land area within each utility's service territory as a control for land-use intensity and population density. According to the U.S. Census Bureau's 2020 definition, an urban area is a densely developed territory that encompasses at least 2000 housing units or a population of 5000 or more. We calculated the share of land classified as urban for each utility. To do this, we used the U.S. Census Bureau's 2020 Urban Areas shapefile, overlaying it with utility service territory polygons and compute the fraction of intersecting land with urban areas.

<sup>4</sup> The DOE's LEAD Tool website writes "Energy burden is defined as the percentage of gross household income spent on energy costs. It is calculated by dividing the average housing energy cost by the average annual household income. A household with 6% or greater energy burden is considered to be a high energy burden household." It references ACEEE (American Council for an Energy Efficient Economy)'s *How High Are Household Energy Burdens? An Assessment of National and Metropolitan Energy Burden across the United States* co-authored by Ariel Dreihobl, Lauren Ross, and Roxana Ayala (2020).

### 4.2.5. Climate conditions

Electricity demand responds strongly to climatic extremes, so we included a weather variable consisting of annual degree-day counts at two thresholds: heating degree days below 15 °C (HDD) and cooling degree days above 30 °C (CDD). Degree days measure how much the daily average temperature deviates from a base level considered comfortable for indoor environments. The greater the deviation from this baseline, the more energy is typically required for space heating or cooling, making degree days a useful proxy for climate-driven energy needs.

We used raster data from the PRISM Climate Group at Oregon State University, which provides daily temperature data across the United States (PRISM Group, n.d.). For each grid cell of 4 km by 4 km size, we compute annual HDD and CDD totals, then aggregate these to utility service territories. Higher HDD values indicate more intensive heating demand, while higher CDD values reflect exposure to extreme heat. Including both allows us to capture how variations in local climate influence residential electricity consumption.

### 4.2.6. Utility characteristics: utility ownership structure

For utility characteristics, no comprehensive database was available for direct use. Therefore, most utility-related data were manually collected from utility websites, while accuracy and relevance for the analysis were ensured through multiple cross-checks among our team members. Specifically, we collected data from 193 utilities across Georgia (91 utilities) and North Carolina (101 utilities). We then classified the utilities into three different types based on their ownership structure: cooperative-owned (COOPs), municipal-owned (MUNIs), and investor-owned (IOUs) utilities. Our dataset includes 121 MUNIs, 67 COOPs, and 4 IOUs.

### 4.2.7. Utility characteristics: electricity pricing rate

This study also incorporated electricity rate into the analysis since price may affect household electricity consumption (e.g., Alberini et al., 2011). We obtained a proprietary dataset from PowerOutage.com that includes electricity rate per kWh for each utility. PowerOutage.com calculates this rate by using state-reported data on total customers, revenues, and sales attributed to specific electric providers (PowerOutage.com, n.d.). To smooth any annual fluctuations in electricity rates, we use the average of five-year electricity rates from 2017 to 2022 for our analysis.

### 4.2.8. Utility characteristics: availability/offerings of energy incentive programs

We tested whether the availability or offerings of utility programs correlates with household energy consumption. If utilities offer programs that are designed to promote energy efficiency, renewable energy use, and EV adoption, their customers might react to these measures by using electricity more efficiently or consuming more electricity due to EV purchase. Conversely, it is possible that utilities provide these programs in response to electricity consumption behaviors of their customers. Since this study did not have panel data, we could not claim any causal relationships between the availability (or offerings) of energy programs and residential electricity use. Additionally, greater program availability or more program offerings do not necessarily imply higher customer uptake. Still, it is worth examining whether availability of programs has connections to energy consumption aggregated at the utility level.

We conducted a systematic manual review of utilities' public websites to compile incentives related to residential electricity use and conservation. In total, we identified 699 programs, 393 in Georgia and 306 in North Carolina. Each program was classified into one of four categories, energy efficiency, renewable energy, electric vehicles, or other—reflecting the distinct ways different incentives may influence household energy use and conservation behaviors. The distribution of programs appears in Table 3. Detailed descriptions of each category

**Table 3**  
Distribution of energy incentive programs.

| Program Type                              | Category      | Incentive type                 | Number of Incentives |
|-------------------------------------------|---------------|--------------------------------|----------------------|
| Energy Efficiency (EE) incentive programs | Financial     | Rebate/Credit                  | 162                  |
|                                           |               | Loan                           | 35                   |
|                                           |               | Discount                       | 8                    |
| Renewable Energy (RE) incentive programs  | Informational |                                | 55                   |
|                                           |               |                                |                      |
|                                           | Financial     | Net metering                   | 68                   |
|                                           |               | Community solar programs       | 22                   |
|                                           |               | Rooftop solar rebate           | 2                    |
|                                           |               | Other                          | 9                    |
|                                           | Other         | Voluntary green power purchase | 42                   |
| Electric Vehicle (EV) incentive programs  | Financial     | Charger installation rebate    | 15                   |
|                                           |               | EV rebate/loan                 | 10                   |
|                                           |               | Discount electricity rate      | 7                    |
|                                           | Other         | Free public charging stations  | 8                    |
|                                           |               |                                |                      |
| Other incentive programs                  |               | Low income assistance          | 82                   |
|                                           |               | Payment assistance             | 71                   |
|                                           |               | Load management programs       | 39                   |
|                                           |               | Other                          | 64                   |
|                                           |               |                                |                      |
| Total                                     |               |                                | 699                  |

follow.

- **Energy efficiency (EE) incentive programs** include all incentives aimed at improving household energy efficiency. In residential electricity context, these programs tend to primarily focus on weatherization. Weatherization involves assessing both home structure and appliances for energy loss and addressing pre-existing inefficiencies by upgrading air sealing, insulation, and incentivizing energy-saving appliances to maximize the highest energy savings-to-investment ratios. Utilities typically provide these programs through financial or informational assistance. Financial assistance can include discount rates for fully electric households, rebates or bill credits for replacing old appliances with Energy Star appliances and providing low-interest loans for energy efficiency improvements. Informational assistance, such as free energy audits, helps households understand how to enhance their energy efficiency.
- **Renewable energy (RE) incentive programs** include all initiatives related to solar, wind, and other renewable energy sources. These programs generally fall into two subtypes: the first *financially benefits households* by offering incentives, such as payments for power generated by personal solar panels or incentives for installing solar panels. The second subtype involves *voluntary contributions* toward a greener energy shift, typically allowing customers to pay extra dollars more for a higher percentage of their energy to come from renewable energy sources, such as community solar programs. Among the 143 renewable energy programs, approximately 101 directly benefit customers (labeled “RE Financial”), while the rest 42 are voluntary contribution programs (labeled “RE Voluntary Contributions”).
- **EV (Electric Vehicle) incentive programs** are another category, specifically focused on incentives to promote electric vehicle adoption or usage. Examples include rebates for home EV charger installation, loans for purchasing EVs, lower electric rates during off-peak hours to encourage EV ownership and charging, and public free charging stations within the utility's service area.
- **Other energy incentive programs** include all remaining programs not classified into the above three categories. Many of these programs focus on billing and payment assistance, such as leveled billing and prepay metering, which help customers manage

electricity expenses by offering predictable monthly payments or flexible prepayment options. Additionally, some programs support community welfare by providing discounts to vulnerable populations, such as senior citizens and low-income households. There are also community support programs designed to help at-risk customers during power outages caused by extreme weather, such as offering portable batteries for customers who rely on electricity for medical needs. Furthermore, utilities also provide tree trimming and Right-of-Way management programs to ensure service reliability by preventing outages due to vegetation interference.

#### 4.3. Spatial aggregation process

To improve the accuracy of spatial aggregation, we applied population-weighted averaging methods when aggregating census tract data to match with utility service boundaries. Fig. 1 illustrates an example of a utility that intersects with five census tracts. To estimate the total population for each census tract and the portion where the utility service area and census tract overlap, we utilized the Oak Ridge National Laboratory's Landscan population raster with a approximately 90m resolution (Weber et al., 2022). For each census tract, we identified a portion that intersects with the utility, referred to as the intersected segment, and a portion that lies outside the utility's boundaries (except for cases where the entire census tract falls within a utility's service territory).

We then used population raster to determine the percentage of the population residing within the intersected segments. This percentage, calculated as the ratio of the population within the intersected segments to the total population of the corresponding census tract, represents a population weight for that census tract. These weights were then applied to all census tract-level variables of interest to determine how many people or households from each census tract should be assigned to the intersected segments. Then, we summed all values from the intersected segments to obtain the total for each utility to compute the final utility-level estimate of that variable. This methodology was to ensure that aggregated metrics accurately reflect the population distribution and demographic characteristics within each utility's boundary, providing a robust foundation for our unique utility-level analysis.

Aside from using a 1-km resolution for weight calculation, our aggregation method is grounded on the accuracies of utility boundaries, meaning more precise boundaries should yield better aggregation results. For this reason, we use a utility boundary shapefile from (PowerOutage.com, 2026) via a restrictive data use agreement; their shapefiles offer greater detail and accuracy compared to the widely used public dataset from the Homeland Infrastructure Foundation-Level Database (HIFLD).

Fig. 2 illustrates these nuances for Georgia and North Carolina, demonstrating how the utility boundaries we use offer an improved precision by excluding areas where utilities do not actually provide service. In these examples, areas colored in blue represent regions serviced by Georgia Power (GA) and Duke Energy Progress (NC) in each state. The white areas within bold boundaries are not serviced by these two large investor-owned utilities. However, the existing HIFLD dataset incorrectly includes all those areas as belonging to the utilities in question, misrepresenting the true extent of utility service boundaries. Therefore, the enhanced identification of utility service areas with the data sourced from PowerOutage.com will make our results more valid.

#### 4.4. Methods

This study analyzes the relationships between household energy consumption patterns and various socioeconomic characteristics, prevalence of energy burden, EV charging stations, utility ownership type, electricity rate, and availability of utility energy programs, as specified in Equation (1). We use stepwise least squares regressions and the unit of analysis is at the utility service boundary level; therefore, our dependent

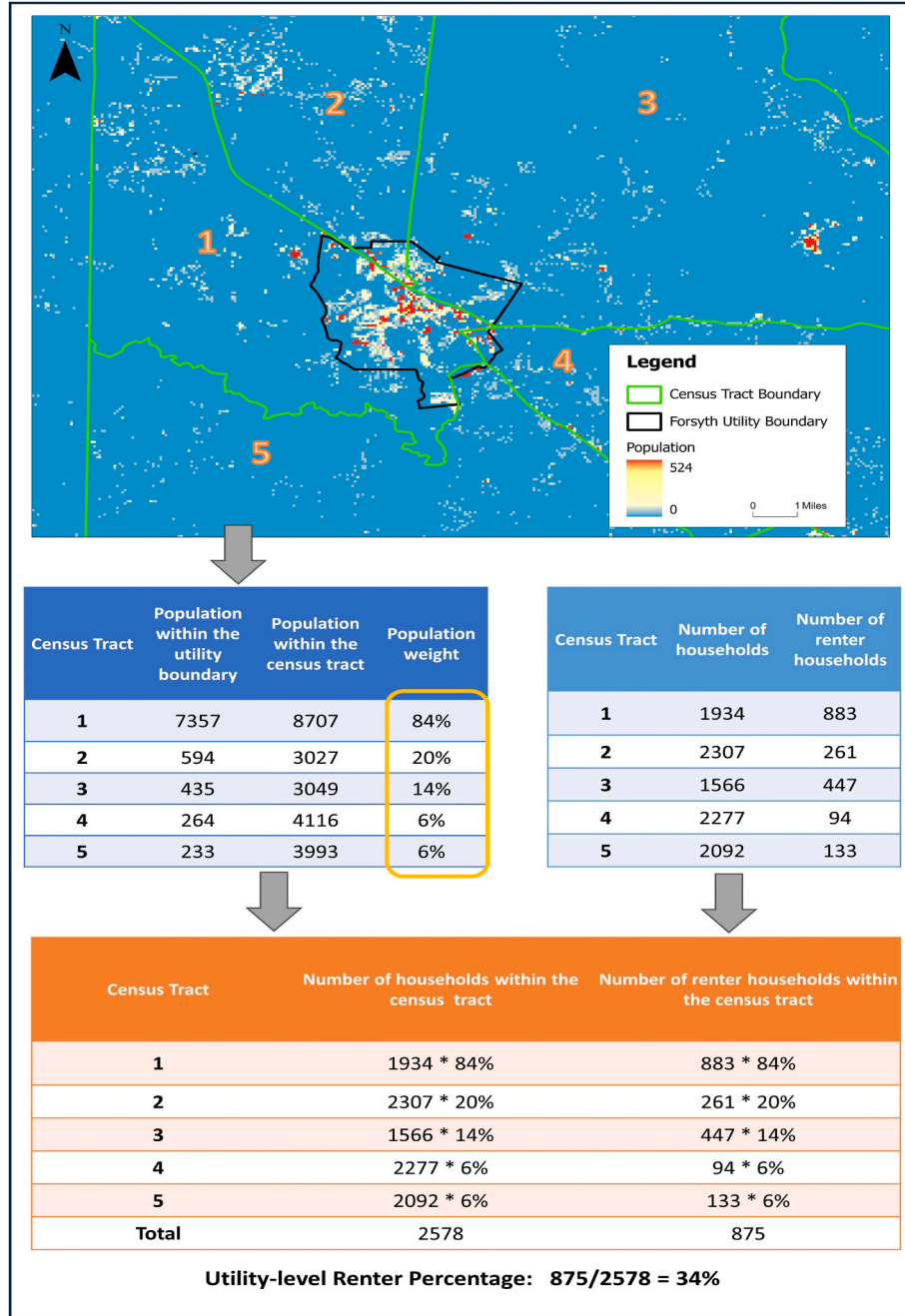


Fig. 1. Example of spatial aggregation.

variable ( $Y_i$ ) is per household electricity use aggregated at the utility level.  $RE_i^d$  and  $RE_i^v$  indicate the number of renewable energy programs that directly benefit consumers and voluntary contribution programs, respectively.  $EE_i$  represents the number of energy efficiency programs and  $EV_i$  is the number of EV related programs.  $X_i^k$  includes socioeconomic characteristics,  $\pi_s$  absorbs unobserved time-invariant heterogeneity within each state.

$$Y_i = \beta_1 RE_i^d + \beta_2 RE_i^v + \beta_3 EE_i + \beta_4 EV_i + \sum_k \gamma_k X_i^k + \pi_s + \varepsilon_i \quad (1)$$

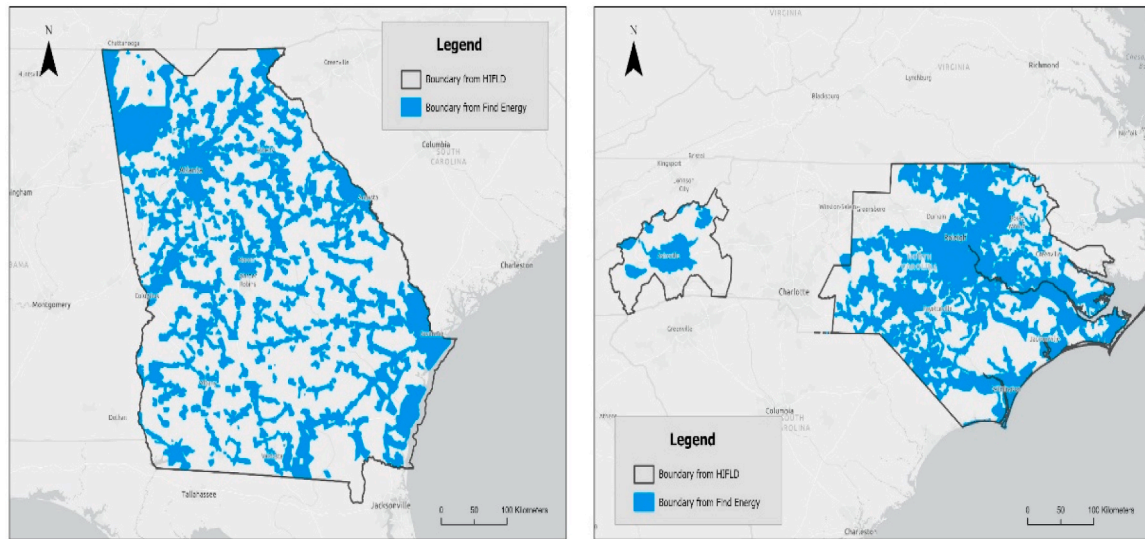
The unit of analysis is electric utility, whereas variables related to socioeconomic characteristics are available initially at the census tract level. We, therefore, develop and apply spatial aggregation process introduced earlier to align the census tract level data with utility service boundaries.

## 5. Results

### 5.1. Descriptive analysis

We first analyze whether the ownership structure of utilities shows different levels of total electricity consumption in the residential sector (RQ3). We conduct an ANOVA test to determine whether electricity consumption is related to utility ownership structure. The p-value is smaller than 0.001, indicating strong statistical significance, meaning that at least one utility ownership structure has significantly different electricity consumption levels. We then perform Tukey's test to identify which groups differ.

The result in Table 4 indicates a statistically significant difference in electricity consumption between cooperative- and municipal-owned utilities with households in the former group consuming more



(a) Georgia Power

(b) Duke Energy Progress

**Fig. 2.** Examples of difference in service boundaries of Georgia power and duke energy progress between HIFLD and PowerOutage.com (formerly FindEnergy).

**Table 4**

Tukey's test comparing average annual household electricity consumption (KWh) between utilities by ownership structure.

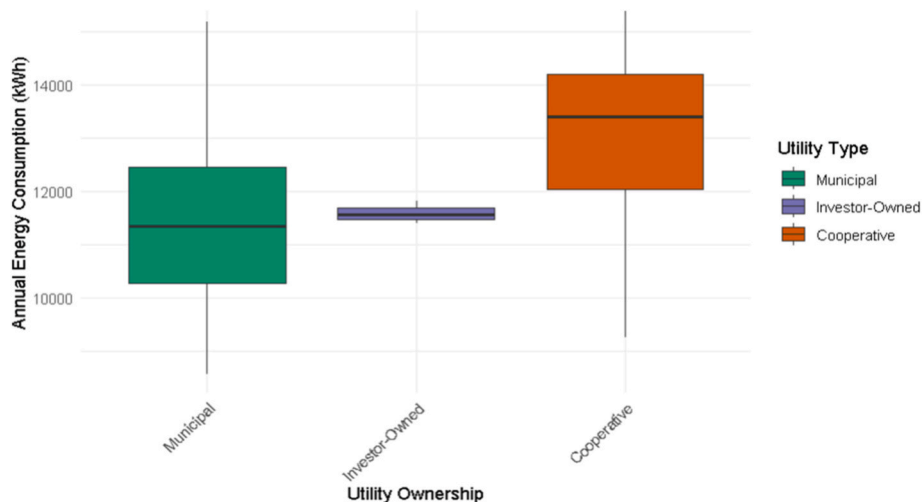
| Comparison                    | Difference | Lower Bound | Upper Bound | Adjusted p-value |
|-------------------------------|------------|-------------|-------------|------------------|
| Cooperative vs Municipal      | 1604.34    | 925.09      | 2283.59     | 0.000            |
| Investor-Owned vs Municipal   | 209.66     | -1827.37    | 2246.69     | 0.968            |
| Investor-Owned vs Cooperative | -1394.68   | -3412.01    | 622.66      | 0.232            |

electricity by as much as 1600 kWh annually on average. This difference is substantial. At the average U.S. residential electricity rate of 16 cents per kWh, it translates to approximately \$256 in additional annual energy costs per household. Similarly, households served by cooperative-owned utilities also use more electricity compared to their counterparts served by investor-owned utilities as much as 1400 kWh on average. However, this difference is not statistically significant. Fig. 3

also visualizes the difference between groups.

Next, we examine the connections between utility's various energy programs and incentives and household electricity consumption (RQ4). Table 5 shows the regression results that only add three types of programs as predictors for annual residential electricity consumption at the utility level. The first column shows that both renewable energy (RE) and EV-support programs (EV) are positive and significantly associated with household electricity consumption. Since the measurement unit for these programs are the number of offerings, one more program available in RE and EV, respectively, is related to 521 and 464 kWh increases in electricity consumption. While EE programs show a negative coefficient as expected, it does not show any meaningful relationship.

In the next column, we dissect RE programs into (1) RE financial incentives which directly benefits customers and (2) RE voluntary contributions which requires voluntary participation from customers often at the expense of more cost. The results show that RE voluntary contribution programs seem to lead to even higher electricity use (669 kWh per one more program) than RE financial incentive programs (438 kWh per one more program), and both are statistically significant. The result on EV programs remain the same while EE programs are not



**Fig. 3.** Annual energy consumption by utility ownership.



**Table 5**

Comparing average annual household electricity consumption by major types of utility energy incentive programs.

| <i>Dependent variable: Average Annual Residential Electricity Consumption per Household (kWh)</i> |                         |                         |
|---------------------------------------------------------------------------------------------------|-------------------------|-------------------------|
| <i>Independent variables</i>                                                                      | (1)                     | (2)                     |
| (Intercept)                                                                                       | 11724.99***<br>(223.57) | 11712.52***<br>(225.49) |
| Energy Efficiency (EE) incentive programs                                                         | −64.35<br>(78.14)       | −55.78<br>(79.94)       |
| Renewable Energy (RE) programs                                                                    | 521.04***<br>(158.22)   |                         |
| RE Financial incentive programs                                                                   |                         | 437.60**<br>(219.71)    |
| RE Voluntary Contributions programs                                                               |                         | 668.82**<br>(312.35)    |
| Electric Vehicle (EV) incentive programs                                                          | 464.47**<br>(224.14)    | 474.80**<br>(225.70)    |
| <i>Observations</i>                                                                               | 105                     | 105                     |
| <i>R2</i>                                                                                         | 0.1983                  | 0.2007                  |
| <i>Adjusted R2</i>                                                                                | 0.1745                  | 0.1688                  |
| <i>AIC</i>                                                                                        | 1538.27                 | 1539.22                 |
| <i>F-statistic</i>                                                                                | 8.328                   | 6.279                   |
| <i>RMSE</i>                                                                                       | 1483                    | 1490                    |

Notes: Robust standard errors in parenthesis; \*\*\*p < 0.01, \*\*p < 0.05, \*p < 0.1.

significant, either. In the next section, we jointly consider all variables introduced earlier.

## 5.2. Multivariate analysis

To examine associations with electricity consumption, we estimate ordinary least squares (OLS) models using the full set of covariates described above. The dependent variable is utility-wide residential electricity consumption per household. Although expressed on a per-household basis, the unit of analysis is the utility service area; therefore, coefficients should be interpreted at the service-area level. Results

**Table 6**

Baseline results.

| <i>Dependent variable: Average Annual Residential Electricity Consumption per Household (kWh)</i> |                       |
|---------------------------------------------------------------------------------------------------|-----------------------|
| <i>Independent variables</i>                                                                      | (1)                   |
| Non-Hispanic White (%)                                                                            | −14.93<br>(11.53)     |
| Higher Education (%)                                                                              | −65.01**<br>(25.91)   |
| Income (\$)                                                                                       | 0.02<br>(0.02)        |
| Single Unit (%)                                                                                   | 28.47<br>(22.51)      |
| Renter (%)                                                                                        | −7.40<br>(18.92)      |
| Median Year Built                                                                                 | 10.68<br>(24.76)      |
| Median # of Room                                                                                  | 1375.91**<br>(570.11) |
| Electric Heating (%)                                                                              | 45.88***<br>(12.80)   |
| State (North Carolina = 1; Georgia = 0)                                                           | 170.66<br>(347.43)    |
| Heating Degree Days                                                                               | 0.40<br>(1.03)        |
| Cooling Degree Days                                                                               | −0.20<br>(13.82)      |
| <i>Observations</i>                                                                               | 105                   |
| <i>R2</i>                                                                                         | 0.590                 |
| <i>Adjusted R2</i>                                                                                | 0.541                 |
| <i>AIC</i>                                                                                        | 1783.2                |
| <i>F-statistic</i>                                                                                | 12.147                |
| <i>RMSE</i>                                                                                       | 1041.65               |

Notes: Robust standard errors in parenthesis; \*\*\*p < 0.01, \*\*p < 0.05, \*p < 0.1.

are reported in Tables 6 and 7.

The baseline results reported in Table 6 only include a set of socioeconomic characteristics and climate variables. There are several notable findings. First, households with higher levels of education are more likely to use less electricity. A one percentage point increase in the population with a bachelor's degree or higher in the utility service territories is associated with a 65 kWh decrease in annual per-household electricity consumption within the service boundary. This relationship may stem from the increased awareness and adoption of energy-efficient practices among more educated populations as the literature suggests (e.g., Wang et al., 2020).

The size of homes also shows a strong relationship with electricity use. Specifically, an increase of one room in the median number of rooms is associated with an increase of approximately 1376 kWh in annual per-household electricity consumption. This finding is consistent with the expectation that larger homes require more energy for space conditioning, lighting, and appliances.

The use of electric heating systems also correlates with higher electricity consumption. A one percentage point increase in the proportion of households using electric heating within the utility service areas is associated with 46 kWh increase in the average annual per-household electricity consumption. This finding aligns with the expectation that electric heating systems require a more significant electricity input, as they replace the fuels provided by natural gas. By contrast and surprisingly, we do not observe any significant relationship with heating or cooling degree days, and these null results do not change even when we include only one variable over the other. One possible reason for such lack of relationship is the climates may not vary substantively by utility service territories across the two adjacent states.

Next, the results reported in Table 7 add additional variables to examine the relationship with specific factors of interest, while still including all the socioeconomic and climate variables appeared in Table 6. For clearer result presentation in the main text, only the newly added variables are shown in the tables; the full results for all variables, including the baseline variables, are provided in Appendix Table A3.

Model 2 in Table 7 builds upon these results by introducing the electricity rate and the share of the population experiencing severe energy burden (i.e., more than 6% of their income spent on energy bills). The electricity rate emerges as a significant factor: every one cent per kWh increase in the average residential electricity rate correlates with a 227 kWh decrease (1.8%) in annual per-household electricity consumption at the utility level. This suggests that higher electricity prices might create an incentive for households to reduce or save usage through behavioral changes or efficiency improvements.

Model 3 expands the analysis by adding utility structural characteristics, including utility type and the share of the service jurisdiction in urban areas. Since many cooperatives serve customers in rural areas, and we want the model to tease out the distinctive role of utility's ownership structure from dynamics of urban-rural split, we control for urbanization in percent of land. Regarding utility ownership structure, model 3 shows a pattern consistent with results from ANOVA and Tukey's post-hoc test: households served by cooperative utilities consume more electricity than those served by municipal-owned utilities. Specifically, cooperative utilities are associated with approximately 1035 kWh higher annual household electricity consumption compared to municipal-owned utilities, on average. This difference likely reflects systematic differences in the customer bases and geographic contexts these utilities serve. Cooperatives frequently serve more rural or remote areas where households tend to live in larger, detached homes that require more energy for heating, cooling, and daily activities. They also face barriers to adopting energy-efficient appliances and infrastructure, further contributing to higher electricity use. In contrast, municipal utilities typically serve more densely populated urban areas with smaller homes or multi-unit dwellings that share walls, and residents have better access to efficient technologies, which helps lower heating and cooling needs.

**Table 7**  
Stepwise results.

| Dependent var: Average Annual Residential Electricity Consumption per Household (kWh) |                           |                            |                            |                            |                            |
|---------------------------------------------------------------------------------------|---------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| Independent variables                                                                 | (2)                       | (3)                        | (4)                        | (5)                        | (6)                        |
| Electricity Rate (\$/kWh)                                                             | −22706.94**<br>(10367.07) | −35352.07***<br>(10383.47) | −32912.86***<br>(10319.33) | −33066.82***<br>(10365.69) | −33379.88***<br>(10372.25) |
| Energy Burdened Population (%)                                                        | 11.78 (12.18)             | 19.58 (12.13)              | 20.59* (11.86)             | 20.50* (11.91)             | 20.75* (11.90)             |
| Utility Type - COOP                                                                   |                           | 1034.57* (471.25)          | 503.10 (514.64)            | 449.78 (525.77)            | 476.54 (518.60)            |
| Utility Type - IOU                                                                    |                           | −62.96 (729.68)            | −509.33 (773.10)           | −538.00 (778.04)           | −507.02 (775.45)           |
| Utility Type - MOU                                                                    | (Omitted)                 | (Omitted)                  | (Omitted)                  | (Omitted)                  | (Omitted)                  |
| Urban Area (%)                                                                        |                           | −10.93* (5.97)             | −10.39* (5.85)             | −10.83* (5.93)             | −10.63* (5.88)             |
| EE Incentive Programs (#)                                                             |                           |                            | −1.43 (61.87)              | −0.07 (62.18)              | −0.23 (62.08)              |
| RE Incentive Programs (#)                                                             |                           |                            | 209.41 (137.08)            |                            | 204.41 (137.68)            |
| EV Incentive Programs (#)                                                             |                           |                            | 297.47* (162.69)           | 303.73* (164.76)           | 310.17* (164.17)           |
| RE Financial Incentive Programs (#)                                                   |                           |                            |                            | 159.28 (165.05)            |                            |
| RE Voluntary Contributions Programs (#)                                               |                           |                            |                            | 346.82 (285.12)            |                            |
| EV Charging Stations (# per 10,000)                                                   |                           |                            |                            |                            | 32.63 (46.76)              |
| Variables in Model 1 included?                                                        | Yes                       | Yes                        | Yes                        | Yes                        | Yes                        |
| Observations                                                                          | 105                       | 105                        | 105                        | 105                        | 105                        |
| R2                                                                                    | 0.612                     | 0.667                      | 0.696                      | 0.697                      | 0.697                      |
| Adjusted R2                                                                           | 0.556                     | 0.606                      | 0.628                      | 0.624                      | 0.625                      |
| AIC                                                                                   | 1781.3                    | 1771.3                     | 1767.8                     | 1769.4                     | 1769.2                     |
| F-statistic                                                                           | 11.036                    | 11.004                     | 10.221                     | 9.645                      | 9.676                      |
| RMSE                                                                                  | 1013.00                   | 938.67                     | 897.17                     | 895.56                     | 894.58                     |

Notes: EE = Energy Efficiency. RE = Renewable Energy. EV = Electric Vehicle. IOU = Investor Owned Utilities. COOP = Cooperative Utilities. MOU = Municipal Utilities. Robust standard errors in parenthesis; \*\*\*p < 0.01, \*\*p < 0.05, \*p < 0.1. For the full results, see Table A3 in the Online Appendix.

This geographic distinction is further captured by the significant negative relationship between the share of urban area within a utility's service territory and household electricity consumption: each one percentage point increase in urban area share predicts a 11 kWh decrease in average annual household electricity use. Together, these findings suggest that both the type of utility and the degree of urbanization shape household electricity demand in similar ways.

Model 4 further expands the analysis by including the number of energy incentive programs offered by each utility: EE, RE, and EV programs. Among these, only the number of EV programs shows a significant association with consumption. Each additional EV program corresponds to an increase of roughly 297 kWh (2.4%) in annual per-household electricity consumption. This may reflect the growing electricity use for EV charging, as these programs encourage the adoption and use of electric vehicles. While EV incentives promote sustainable transportation, they seem to predict higher household electricity usage. By contrast, the numbers of EE and RE programs do not show a significant relationship with average household consumption at the utility level. This result may suggest that while these programs aim to reduce consumption or promote energy savings, their effects are either gradual, spread across many sectors, or even potentially offset by rebound effects and new sources of demand.

Incorporating these program variables also leads to notable changes in the significance of several other predictors. First, the percentage of energy-burdened households becomes significant in Model 4, indicating that utilities with a greater share of households spending a high proportion of income on energy tend to have higher per-household consumption. Because energy burden is partly constructed from energy expenditures, its positive association with electricity consumption should not be interpreted as an independent behavioral mechanism in isolation. The more policy-relevant result is that the association emerges in models that control for median household income and other service-territory characteristics. This suggests that high-burden utility territories may face structural inefficiencies, such as inefficient housing stock, limited access to efficient appliances, or constrained ability to invest in retrofits.

At the same time, the utility type-coop becomes insignificant when program variables are included, suggesting that differences in household energy consumption previously attributed to utility governance structure may be partly explained by the types and numbers of incentive

programs offered, as well as by overlapping demographic factors.

Model 5 builds on this by disaggregating the RE programs into two subtypes: RE voluntary contribution and financial incentive programs. This distinction aims to examine whether different forms of RE offerings have varied impacts on household electricity consumption. The results show that neither of the RE program counts have a significant association with average household electricity use, mirroring the earlier finding in Model 4 when only using the total count of RE programs. This suggests that simply offering more renewable energy program options, regardless of type, may not bear a clear direct effect on reducing household electricity consumption at the utility level.

Model 6 further tests the robustness of the significant relationship between the number of EV programs and household electricity consumption by adding a control variable: the number of public EV charging stations per 10,000 people. This variable serves as a proxy for existing local EV adoption rates, based on the assumption that areas with more widespread EV ownership will require more public charging infrastructure. By including this control, the model tests whether the association between EV programs and household electricity use remains after accounting for local charging infrastructure. The results show that an additional EV program remain significantly associated with 2.5% higher per-household electricity consumption. This pattern is consistent with the possibility that EV program offerings are linked to greater EV-related residential charging demand, although the cross-sectional design does not allow causal identification.

## 6. Limitations

Before we discuss our research findings and policy recommendations, it is important to acknowledge some of the limitations in the current study. Among several variables we examine, one of our interest is the relationships between household energy consumption and the availability of utility programs within service territories (i.e., RQ4). It addresses a gap in the literature that has mostly focused on county-level household energy use rather than utility-level patterns. Due to the absence of centralized databases for utility programs, we relied on manual collection of program information from each individual website across 193 utilities in Georgia and North Carolina. While centralized repositories such as the Database of State Incentives for Renewables & Efficiency (DSIRE) exist, their coverage is incomplete with a

predominant focus on large IOUs. For example, in our comprehensive data collection, we identified 259 energy efficiency programs across Georgia utilities, most of which are absent from DSIRE's database, specifically those offered by municipal and cooperative utilities. However, the greater program availability does not necessarily imply higher customer uptake, and this remains one of the limitations of the current study.

Additionally, our analysis is limited to a cross-sectional research design because most utilities do not provide historical information about when programs were initiated, modified, or terminated. This prevents us from conducting panel analysis that could better establish causal relationships between program availability or offerings and energy consumption patterns. Additionally, utilities may inconsistently update their websites, potentially leading to measurement error where terminated programs appear active or where program details (such as incentive amounts or eligibility criteria) have changed without documentation. In addition to the availability of utilities' energy programs, future research could leverage program uptake information if such data is publicly available from certain or a limited number of utilities or can be obtained via a non-disclosure agreement.

Finally, our analysis is limited to two Southeastern states which restricts the generalizability of our findings to other regions with different climate conditions, regulatory environments, or utility market structures. The Southeastern U.S. has particular characteristics, including higher electric heating penetration and different cooling patterns, that may not be representative of national electricity consumption patterns. While we identified 193 utilities across both states, our final analysis includes 105 utilities due to missing data in the EIA-861 database. Although the excluded utilities represent only 4% of the total population (see the online appendix), this selection may introduce bias if smaller utilities have systematically different program offerings or consumption patterns.

Despite these limitations, our study makes several important contributions to the literature. This study is the first systematic analysis of utility-level energy consumption patterns that considers the comprehensive landscape of utility programs beyond large IOUs, including the municipalities and cooperatives that serve significant portions of residential customers in both urban and rural areas. As will be discussed in the next section, our findings offer insights for policymakers and utility planners seeking to understand how different types of energy incentive programs relate to residential electricity usage patterns.

## 7. Discussion and policy recommendations

This study yields five key findings. First, utility type and geographic context (e.g., rural versus urban) are significantly related to residential electricity consumption. Households served by cooperative utilities and located in rural areas tend to consume more electricity, largely reflecting differences in housing size and building type.

Second, socioeconomic and structural factors, including electricity rates, education levels, home size, and heating type, are strong predictors of household energy use. Higher electricity prices and greater educational attainment are associated with lower consumption, while larger homes and electric heating correspond to higher use. The prevalence of energy burden is also positively associated with electricity consumption, indicating that low-to moderate-income households facing high energy costs often reside in less energy-efficient homes.

Third, EV incentive programs are positively associated with residential electricity demand, even after controlling for the density of public charging stations, consistent with the possibility that EV programs are associated with greater EV adoption and at-home charging demand, although the cross-sectional design does not allow causal identification. In contrast, the number of energy efficiency (EE) and renewable energy (RE) programs show no significant relationship with household electricity use once other covariates are considered. When examined in isolation, however, both EV and RE programs, whether

financial or voluntary, are linked to higher consumption, while EE programs consistently show no measurable association.

These program results should be interpreted as associations between program availability and utility-level electricity consumption, not as estimates of program effectiveness. Our program variables measure whether utilities offer EE, RE, and EV programs, but they do not capture customer uptake, incentive level, program spending, eligibility stringency, participation rates, or realized energy savings. Therefore, the null association for EE programs should not be interpreted as evidence that EE programs are ineffective. A utility with one large, well-funded, high-participation efficiency program may produce larger consumption changes than a utility with several small programs. Program impacts may also be gradual, diluted across the full utility customer base, or offset by rebound effects. Similarly, positive associations for EV and RE programs may reflect electrification, at-home EV charging, distributed generation participation, or other electricity-using technologies, but the cross-sectional design does not allow causal interpretation. Future research should distinguish program availability from uptake, spending, incentive size, participation rates, and realized savings.

Next, the significance of household and demographic characteristics (such as education, home size, and electric heating) diminishes once utility ownership and programmatic variables are included, suggesting that institutional and program factors partly mediate these relationships. Interestingly, heating and cooling degree days do not significantly predict electricity use, while rural households consistently consume more electricity than urban counterparts.

Finally, the significance of electricity rate remains robust across all models, confirming the importance of price signals in shaping consumption. The share of energy-burdened households becomes a significant predictor after accounting for programmatic factors, underscoring structural inequities in energy use. Cooperative utilities initially appear to have higher household electricity use, but this association becomes statistically insignificant once program offerings and demographic factors are included. Taken together, these findings suggest that utility program portfolios and service-territory characteristics help account for variations in residential electricity use that might otherwise be attributed to ownership structure alone.

The negative association between electricity rates and consumption should also be interpreted carefully. Lower electricity use in higher-price utility territories may reflect efficiency improvements, behavioral conservation, or greater responsiveness to price signals. However, in high-burden communities, reduced consumption may also reflect energy-limiting behaviors, including delaying or avoiding heating, cooling, or other essential energy services. This distinction matters because lower electricity consumption is not necessarily welfare-improving if households are reducing use below levels needed for thermal comfort, health, or safety. Evidence from air-conditioning rebate programs similarly suggests that efficiency interventions can reduce energy use and bills while also changing the ability of low-income households to use cooling services, meaning that consumption reduction alone is an incomplete measure of program success (Wade et al., 2025).

Based on these findings, we suggest the following policy recommendations:

1. *Design Smarter EV, Renewable, and Efficiency Programs to Manage Load Growth.* Our analysis shows that voluntary RE contributions programs (i.e., voluntary green power purchase), RE financial incentives (e.g., net metering, community solar programs, and rooftop solar rebate), and EV incentives are linked to increased household electricity consumption when we do not consider other factors. In contrast, EE programs show no consistent relationship. Future research with program-specific data could examine the relationships for disaggregated EE programs. After accounting for other factors, EV incentives remain the strongest predictor of higher energy use, emphasizing the need for program redesign to avoid undermining conservation efforts. As more EV incentive programs proliferate, they

will inevitably increase EV adoption and electricity consumption. Additionally, depending on the time of the increase in electricity usage likely from more EVs can be problematic than mere increase in electricity consumption. While our data cannot disentangle the timing of this increase, if it is primarily focused on peak time, it can lead to stress on the grid and increased capital cost for utilities, as utilities are required to build additional generation capacities based on peak demand (Joskow and Jean, 2007; Blonz, 2022).

To mitigate this surge in residential electricity use, utilities can pair EV and RE incentives with mandatory demand response participation or time-of-use pricing to reduce peak demand and curb unnecessary load growth. Utilities can consider integrating behavioral nudges, real-time usage dashboards, or consumer education campaigns to promote energy-saving behaviors among incentive recipients. Policymakers could require utilities to disaggregate and report EE program impacts separately to clarify their effectiveness and enable performance benchmarking. Additionally, while mostly at pilot stages, when vehicle-to-grid (V2G) or grid-to-vehicle (G2V) programs become available, policymakers and utilities can benefit from shaving the load and reducing peak demand pressures (Brown and Soni, 2019). Finally, the use of integrated program evaluation frameworks that examine the combined and interactive effects of EE, RE, and EV programs on total system load could be further considered.

2. *Use Customer Segmentation to Deliver High-Impact, Equity-Informed Efficiency Interventions.* Household-level factors, such as income, education, and home size, are strongly correlated with higher energy consumption. Programs that treat customers as a homogeneous group risk misallocating resources and failing to meet efficiency goals. We, therefore, provide the following recommendations: (1) Implement targeted outreach and tailored program offerings based on demographic and usage segmentation to ensure program relevance and effectiveness; (2) Expand equity-focused initiatives for under-resourced households, while also strategically engaging high-usage, affluent customers to maximize system-wide efficiency gains; and (3) Integrate segmentation strategies into customer relationship management systems to enable proactive, data-informed engagement and program delivery.
3. *Leverage Dynamic Pricing Strategies to Shape Demand and Manage Grid Load.* Electricity pricing emerged as the most consistent and powerful driver of lower consumption in our models. Customers respond predictably to price signals, when rates vary by time or usage level. Utilities can expand the deployment of dynamic pricing structures such as time-of-use (TOU), real-time pricing (RTP), and critical peak pricing (CPP), with a targeted focus on EV owners and large-load customers. Additionally, utilities can align rate designs with grid conditions and peak demand periods to reduce system strain and avoid costly infrastructure expansion (Matisoff et al., 2020; Blonz, 2022). Introducing incentives for off-peak EV charging, including rebates for smart chargers and participation in managed charging programs can complement these efforts. Finally, utilities should ensure transparent and accessible communication about pricing structures, usage impacts, and cost-saving opportunities to enhance customer awareness and acceptance. These dynamic pricing scheme and transparent communication can be implemented at low-cost due to widely adopted smart meters (Joskow & Wolfram, 2012), especially in Georgia and North Carolina where the advanced metering infrastructure penetration rate exceeds 90% (EIA, 2024).
4. *Integrate Demand Response and Load Management into EV Program Planning.* Utilities with a greater number of EV programs tend to experience higher aggregate consumption. This trend reinforces the need to embed grid-aware charging and load management into EV deployment strategies, for example, (1) by requiring the installation of smart charging infrastructure that is capable of dynamic load

shifting based on real-time grid conditions and customer preference; (2) incorporate vehicle-to-grid (V2G) and managed charging capabilities into public EV infrastructure and residential rebate programs; (3) integrate EV adoption projections into utility Integrated Resource Plans (IRP) to align load forecasts with capital planning and grid investments; and (4) foster public-private partnerships to scale resilient, interoperable EV infrastructure while optimizing grid integration and community access.

As electrification accelerates through transportation and distributed energy resources, utility planning must transition from siloed programs to systems-level coordination. Understanding how EV, RE, and EE programs interact is essential for ensuring that they complement, rather than conflict with, grid reliability and decarbonization goals. Strategic integration of demand response, customer segmentation, and dynamic pricing will be vital to shaping a flexible, equitable, and future-ready energy system.

## 8. Conclusions

This study examined how residential electricity consumption across utilities in Georgia and North Carolina relates to socioeconomic, structural, climatic, and programmatic factors at the utility level. By aggregating diverse data sources, including utility governance characteristics, electricity pricing, program offerings, and demographic and housing variables, this analysis moves beyond the conventional household or county-level perspectives to uncover systemic patterns of demand embedded in utility structures. Our findings show that electricity use correlates not only by household characteristics but also by the institutional and programmatic contexts in which utilities operate.

The results reveal that higher electricity prices and greater educational attainment are consistently associated with lower average household electricity use, while larger homes and greater reliance on electric heating correspond with higher demand. Cooperative utilities tend to serve households with higher per-capita consumption, reflecting their rural service territories and larger home sizes, though these structural effects diminish once energy programs and demographic factors are considered. Among programmatic interventions, the number of EV incentive programs stands out as a robust predictor of increased residential electricity demand, even after accounting for local charging infrastructure. By contrast, energy efficiency (EE) and renewable energy (RE) incentive programs show no significant relationship with household electricity use, suggesting that their impacts may be gradual, offset by rebound effects, or diluted across multiple sectors.

These results carry several implications for both policy and practice. As electrification expands through transportation and distributed generation, utilities face the challenge of promoting sustainability without exacerbating demand pressures or inequities. Aligning EV and RE incentives with demand response mechanisms and time-of-use pricing could help manage load growth while maintaining decarbonization momentum. At the same time, equity-informed targeting of efficiency interventions remains essential for addressing the structural causes of high energy burden in the Southeastern U.S., where energy poverty and inefficient housing remain widespread.

Future research should extend this analysis temporally and geographically to capture causal relationships and regional heterogeneity in program effectiveness. Incorporating program uptake data, panel designs, and household-level behavioral responses would allow a deeper understanding of how utilities' interventions translate into measurable efficiency gains. As utilities increasingly serve as implementing agents of state and federal decarbonization policies, assessing their collective role through utility-level metrics offers a promising direction for evaluating the performance, equity, and resilience of the U.S. energy transition.



## CRediT authorship contribution statement

**Geyu Lyu:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **John Kim:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft, Writing – review & editing. **Yu Wang:** Conceptualization, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Chien-fei Chen:** Conceptualization, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. **Brian An:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2026.115413>.

## Data availability

Data will be made available on request.

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